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Propability 1 Final exam

1.a. 200 times 40TT, 60HT, 54TH, 46HH

	O_i	E_i	$O_i - E_i$	$(O_i - E_i)^2$	$\frac{(O_i - E_i)^2}{E}$
HH	46	50	-4	16	0.32
HT	60	50	10	100	2
TH	54	50	4	16	0.32
TT	40	50	-10	100	2

Sum = 4.64

degree of freedom = 4 - 1 = 3

$\chi^2 (3, 0.05) = 7.81$

$4.64 < 7.81$

So we can conclude that the coin is fair

Q.2. a. $f(x) = \frac{3}{8}x^2$

$$\int \frac{3}{8}x^2\,dx = \frac{3}{8}\int x^2\,dx$$

$$= \frac{3}{8}\cdot\frac{x^3}{3} + C$$

$$= \frac{x^3}{8} + C$$

i. $\int_2^1 \frac{3}{8}x^2\,dx$

$$= \left[\frac{x^3}{8}\right]_1^2$$

$$= \frac{8}{8} - \frac{1}{8}$$

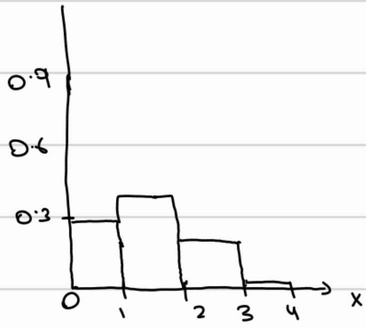
$$= \frac{7}{8}$$

ii. $\int_0^x \frac{3}{8}x^2\,dx$

$$= \frac{x^3}{8}$$

so $F(x) = \begin{cases} 0, & x < 0 \\ \frac{x^3}{8}, & 1 \le x \le 2 \\ 1, & x > 2 \end{cases}$

Q.3. a. $P(T) = \frac{2}{3}$ $P(H) = \frac{1}{3}$



i. X = no. of H

X	0	1	2	3
P	$\overset{0}{\underset{3}{C}} \cdot \left(\frac{1}{3}\right)^{0}\left(\frac{2}{3}\right)^{3}$	$\overset{1}{\underset{3}{C}}\left(\frac{1}{3}\right)^{1}\left(\frac{2}{3}\right)^{2}$	$\overset{2}{\underset{3}{C}}\left(\frac{1}{3}\right)^{2}\left(\frac{2}{3}\right)^{1}$	$\overset{3}{\underset{3}{C}}\left(\frac{1}{3}\right)^{3}\left(\frac{2}{3}\right)^{0}$
P	$\frac{8}{27}$	$\frac{4}{9}$	$\frac{2}{9}$	$\frac{1}{27}$

ii. $P(a \geq 1) = P(1) + P(2) + P(3)$

$= \frac{4}{9} + \frac{2}{9} + \frac{1}{27}$

$= \frac{19}{27}$

4.a. $\mu = 1000$ between 900 and 1300

$\sigma = 200$

$$\frac{900-1000}{200} = -0.5\ z \qquad \frac{1300-1000}{200} = 1.5$$

Using the table $p(x \leq 900) = 0.309$ $P(x \geq 1300) =$

$$p(1300 \geq x \geq 900) = 1 - P(x \leq 900) - P(x \geq 1300$$
$$= 1 - 0.309 - 0.067$$
$$= 0.624$$

Q.5 a. $P(H) = \frac{2}{3}$ $P(T) = \frac{1}{3}$

X = no of heads

X	0	1	2
$P(X)$	${}^2C_0 \cdot \frac{2}{3}^0 \cdot \frac{1}{3}$	${}^2C_1 \cdot \frac{2}{3}^1 \cdot \frac{1}{3}$	${}^2C_2 \cdot \frac{2}{3}^2 \cdot \frac{1}{3}^0$
$P(X)$	$\frac{1}{9}$	$\frac{4}{9}$	$\frac{4}{9}$

$$E[X] = \sum a_i \, P(a_i)$$
$$= 0 \cdot \frac{1}{9} + 1 \cdot \frac{4}{9} + 2 \cdot \frac{4}{9}$$
$$= \frac{4}{3}$$

$$E[X^2] = 0^2 \cdot \frac{1}{9} + 1^2 \cdot \frac{4}{9} + 2^2 \cdot \frac{4}{9}$$
$$= \frac{4}{9} + 4\,\frac{4}{9}$$
$$= \frac{20}{9}$$

$$Var(X) = E(X^2) - E(X)^2$$
$$= \frac{20}{9} - \left(\frac{4}{3}\right)^2$$
$$= \frac{4}{9}$$